Innovation & Entrepreneurship for Engineers

Dr Shankar

IIT Madras

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Innovation & Entrepreneurship for Engineers

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  Mahindra & Mahindra
  Technology Innovation & IP
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- TEDx Speaker
- Prolific Inventor 40+ Patents



Innovation & Entrepreneurship for Engineers

Module 1

Engineers as Innovators

#1. What is Innovation ?

#2. Innovation through S & T

Module 2

Problem Discovery

#3. Finding Problems worth solving

#4. Customer Discovery & Design Thinking

Module 3

Creativity & Innovation

#5. Innovative Thinking – TRIZ, FLOW, CREATE

#6. Enhancing Innovation – AI - Collaboration

Module 4

Deep-Tech Entrepreneurship

#7. Deep – Tech Ventures

#8. Exponential & Systems Thinking

Module 5

Building a Startup

#9. Economics of Innovation & Entrepreneurship

#10. Business Models

#11. Intellectual Property

Module 6

Prototyping & Presentation

#12. Rapid Prototyping

#13. Pitch Your Idea

#14. Demo Day

Innovation Equation

Innovation = Idea + New Value + Path to Market

Innovation is the specific instrument of entrepreneurship.
It is the act that endows resources with a new capacity to create wealth – Peter Drucker

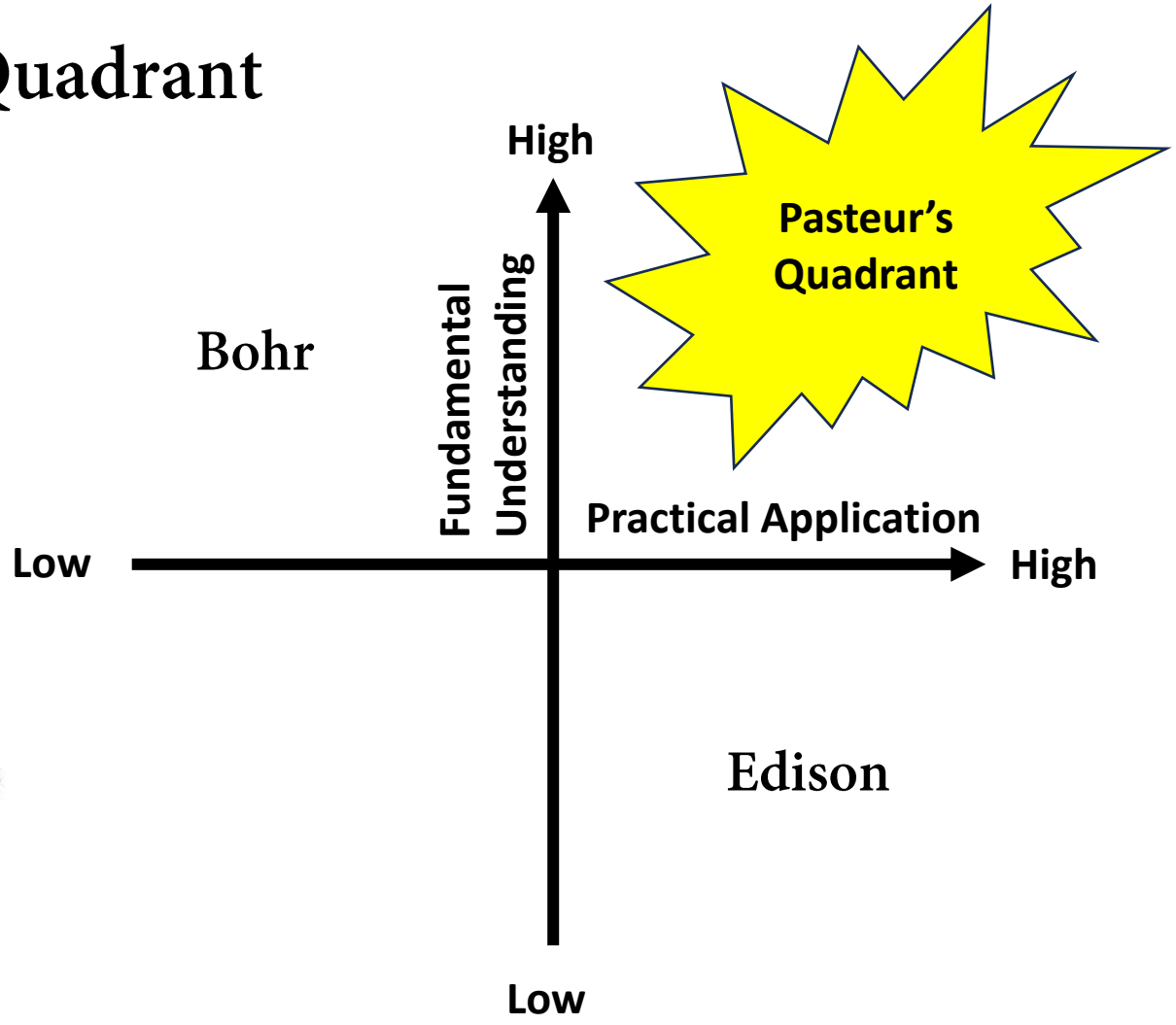
From Idea to Innovation

Idea → **Idea - Problem** → **Desirable Idea**
Scalable Idea
Sustainable Idea → **Feasible Idea**
Patentable Idea → **Viable Idea that creates value to the Customer**



Pasteur's Quadrant



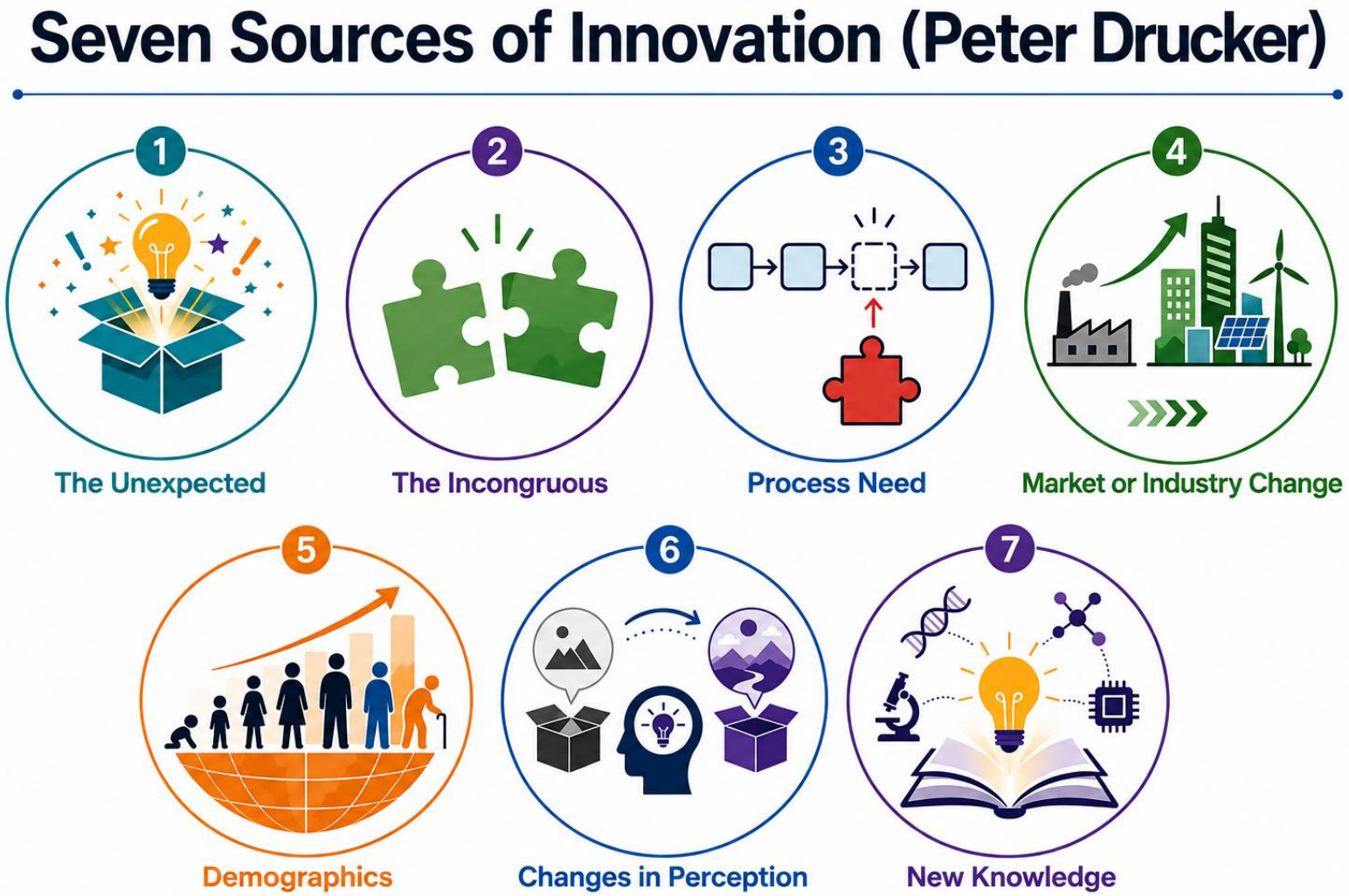


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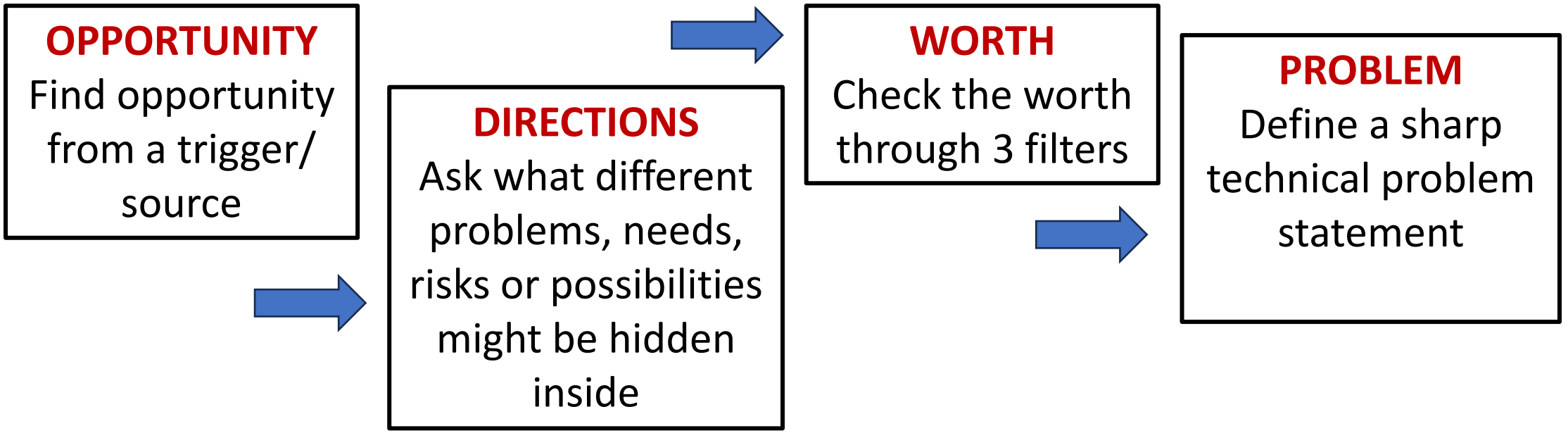
Pasteur's Questions

- Why does this happen – Scientific Curiosity
- What fundamental mechanism is responsible – Deep Understanding
- Can that understanding solve an important problem ?
- Can the solution be translated into practice ?

Why does this happen – and can understanding it help us to solve an Important problem ?


Seven Sources of Innovation (Peter Drucker)
1
The Unexpected
2
The Incongruous
3
Process Need
4
Market or Industry Change
5
Demographics
6
Changes in Perception
7
New Knowledge

How to define a PROBLEM STATEMENT from an OPPORTUNITY



Trigger – Question Mapping

Sources of Innovation	One simple question to ask
Unexpected Event	What new thing does this surprise make us look at?
Incongruity	What is not matching here?
Process Need	Which part of this process needs to work better?
Industry / Market Change	Because the market is changing, what will people or businesses need differently?
Demographic Change	Because people are changing, what will they need differently?
Change in Perception	Because people now think differently, what will they want differently?
New Knowledge	Where can this new ability be useful?

Checking which direction statement is worth pursuing – three filter check



Customer Discovery & Design Thinking

Design Thinking

Dr Shankar

IIT Madras

04 / 05 Sept 2026

“Engineers are trained to solve problems.

Today, I want to teach you something more important:

how to ensure that we are solving the right problem

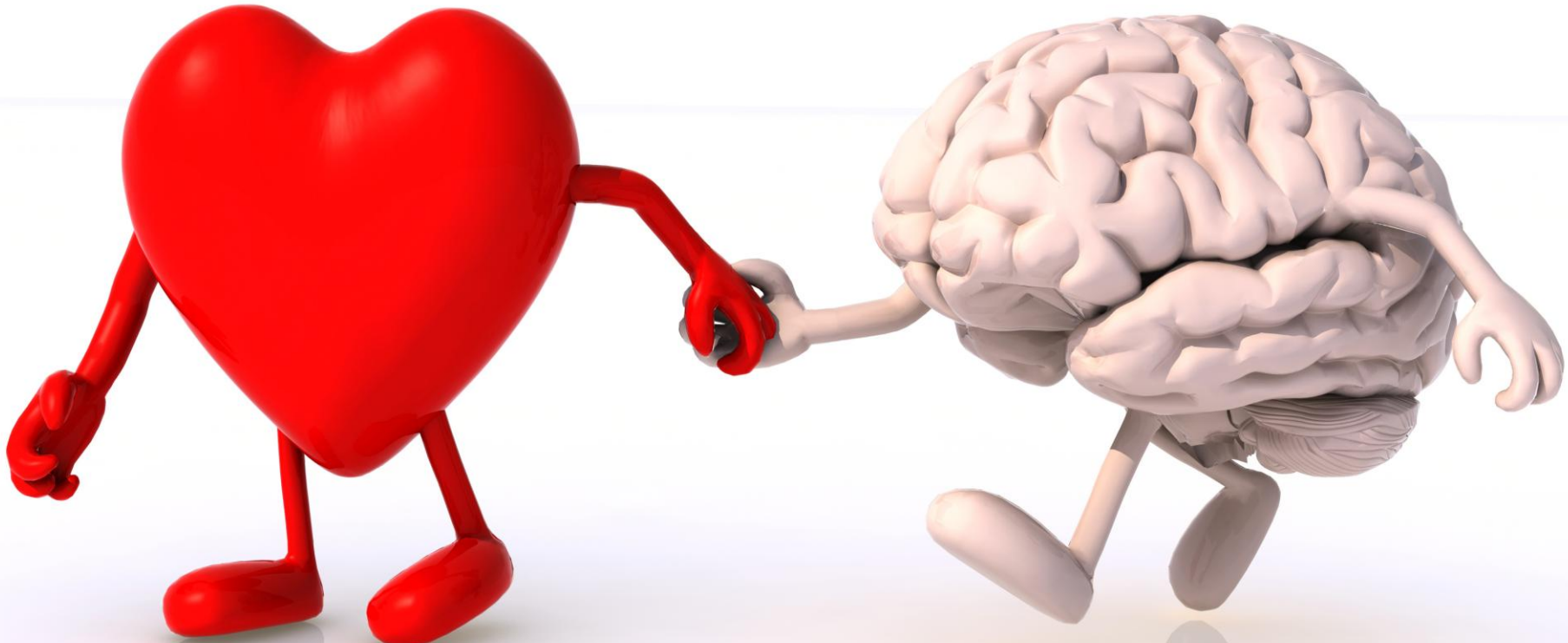


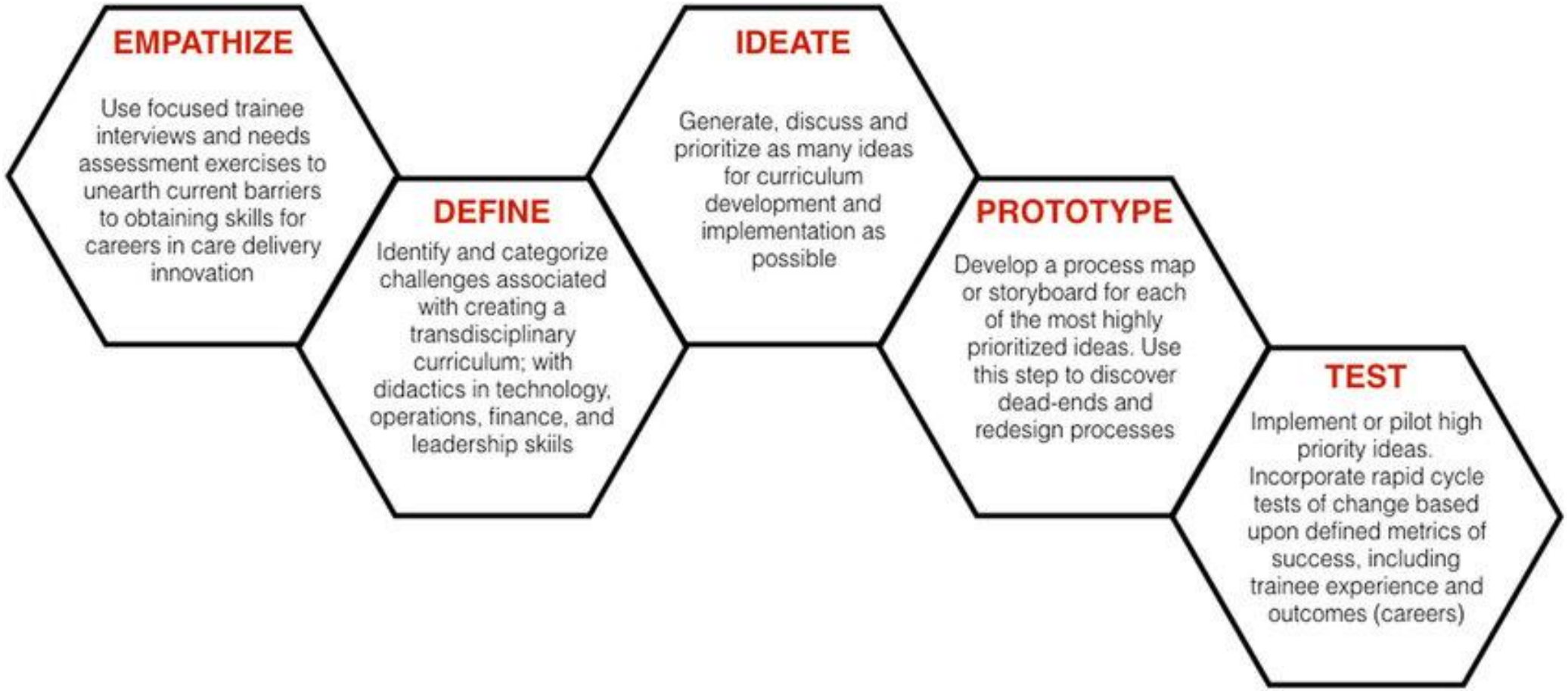
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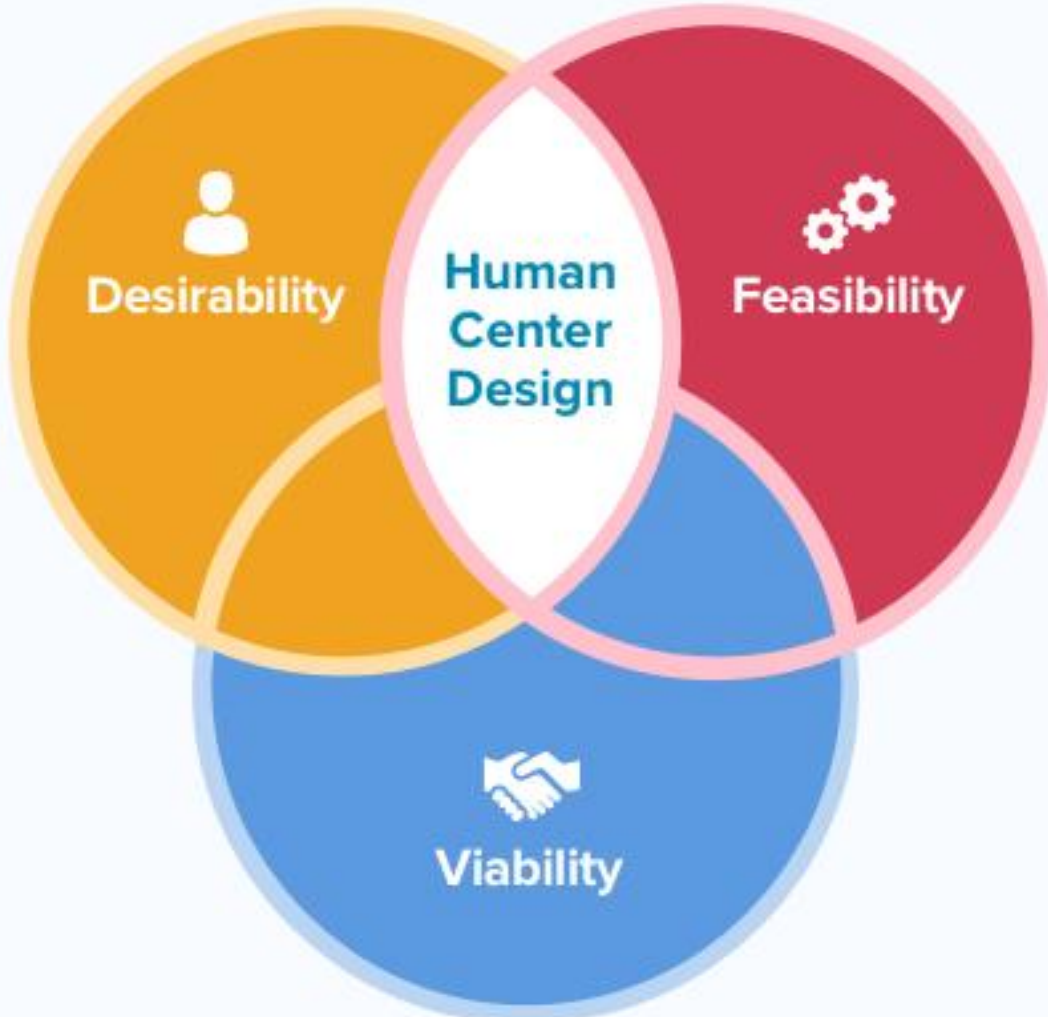
Empathize

Human Centered Design




EMPATHIZE
Use focused trainee interviews and needs assessment exercises to unearth current barriers to obtaining skills for careers in care delivery innovation
DEFINE
Identify and categorize challenges associated with creating a transdisciplinary curriculum; with didactics in technology, operations, finance, and leadership skiils
IDEATE
Generate, discuss and prioritize as many ideas for curriculum development and implementation as possible
PROTOTYPE
Develop a process map or storyboard for each of the most highly prioritized ideas. Use this step to discover dead-ends and redesign processes
TEST
Implement or pilot high priority ideas. Incorporate rapid cycle tests of change based upon defined metrics of success, including trainee experience and outcomes (careers)

Human-Centered Design Principles



Design Thinking Is Not a 7-Step Assembly Line — It's an Iterative Learning Loop

1. Frame a Question
2. Gather Inspiration
3. Synthesize for Action
4. Generate Ideas
5. Make Ideas Tangible
6. Test to Learn
7. Share the Story

Teams repeatedly diverge to explore possibilities and converge to make choices.

Testing may send them back to redefine the problem, gather more insights, or generate entirely new ideas.

Today's Plan

- **Part 1** — Are We Solving the Right Problem?

  Steps 1–3: Frame a Question, Gather Inspiration,

  Synthesize for Action

- **Part 2** — How Do We Create Solutions Nobody Has Thought Of?

  Steps 4–5: Generate Ideas, Make Ideas Tangible

- **Part 3** — How Do We Know It Works — and

  How Do We Make the World Care?

  Steps 6–7: Test to Learn, Share the Story

RUNNING CASES

Aravind Eye Care — redesigning eye-care delivery in India.

Airbnb — the founders' early struggle to get people to trust and book a stranger's home.

Plus: your own team's campus challenge, carried through all three parts.

Today's Challenge — carried across all three parts

How might we make everyday movement across

the large IITM campus effortless, sustainable and inclusive?

This is the through-line challenge your team will frame, ideate, prototype, and later pitch across today's three mini-sessions —

One Challenge, Every Engineering Discipline

Discipline	How They Contribute
Mechanical	Vehicles, mechanisms, ergonomics
Electrical	Energy systems, charging, sensing
Electronics	Sensors, embedded systems, IoT
Computer Science	Algorithms, platforms, AI
Civil	Infrastructure and systems
Chemical	Materials and sustainability
Aerospace	Systems thinking and optimization

Are We Solving the Right Problem?

Innovation frequently fails not because we cannot solve the problem —

but because we solve the wrong problem.

"Go out. See people. Discover what the real problem is."

Now imagine I ask a professor, an office staff, a delivery rider,

a B.Tech student, a Ph.D student, a bus driver and
an ambulance driver the same question.

Will they give the same answer? No.

A problem is experienced by a person, in a context. This is where Design Thinking begins.

Step 1 · Frame a Question

- **Bad framing:** “How can we build an affordable electric scooter for students to move in the campus?” — the solution is already assumed.
- **Better framing:** “How might we help students to travel inside the campus conveniently?”
- **Even better:** “How might we help a B.Tech student (male / female) to mange their diverse interests and move across different parts of the campus easily, efficiently, safely and inclusively ?

Case 1 — Aravind Eye Care

- Millions of people suffer from preventable blindness because cataract surgery is unavailable or unaffordable.
- An engineer's first instinct: build better surgical equipment. Develop cheaper instruments. Automate surgery.

BUT WHAT IF THE FUNDAMENTAL PROBLEM ISN'T THE SURGICAL TECHNOLOGY?

How might we deliver high-quality eye care to millions of people — including people who cannot afford conventional medical treatment?



https://www.ted.com/talks/thulasiraj_ravilla_how_low_cost_eye_care_can_be_world_class?utm_campaign=tedspread&utm_medium=referral&utm_source=tedcomshare

This reframing expands the solution space

"Improve surgery."

"Redesign the entire system of delivering eye care."

Step 2 · Gather Inspiration

Method	What It Reveals
Observe	What do people actually do?
Listen	What do they say?
Immerse	What does their experience feel like?
Look for extremes	Who experiences the problem most intensely?

Case: Airbnb — Don't Optimize the Spreadsheet, Go Meet the Human

DEEPER PRINCIPLE

Don't optimize the spreadsheet.

Go and meet the human.

- Bookings were struggling in Airbnb's early years.
- A conventional tech response: improve the search algorithms, or spend more on digital advertising.
- Instead, the founders went directly to users — they visited hosts and looked closely at the actual experience.
- Key insight: many listings had poor-quality photographs. The online experience failed to communicate what the accommodation actually felt like.
- The team experimented with improving photography and the presentation of listings.

A Live Classroom Experiment

- How many of you have ordered food online?

  What is the problem with food delivery?

- How many of you have actually watched a delivery person work for three hours?

  Probably none.

EXPECT

Delivery time.

Cost.

Packaging.

Food quality.

Step 3 · Synthesize for Action

- Imagine you interview 20 students about hostel food. You collect: 200 quotes, 100 observations, 50 photographs, 30 complaints.
- As Engineering students, we often want to immediately say: “Let's build an app.”
- Data is not insight. You now need to sort, cluster, compare, look for patterns, find contradictions, and identify opportunities.

WHAT NEXT?

Sort

Cluster

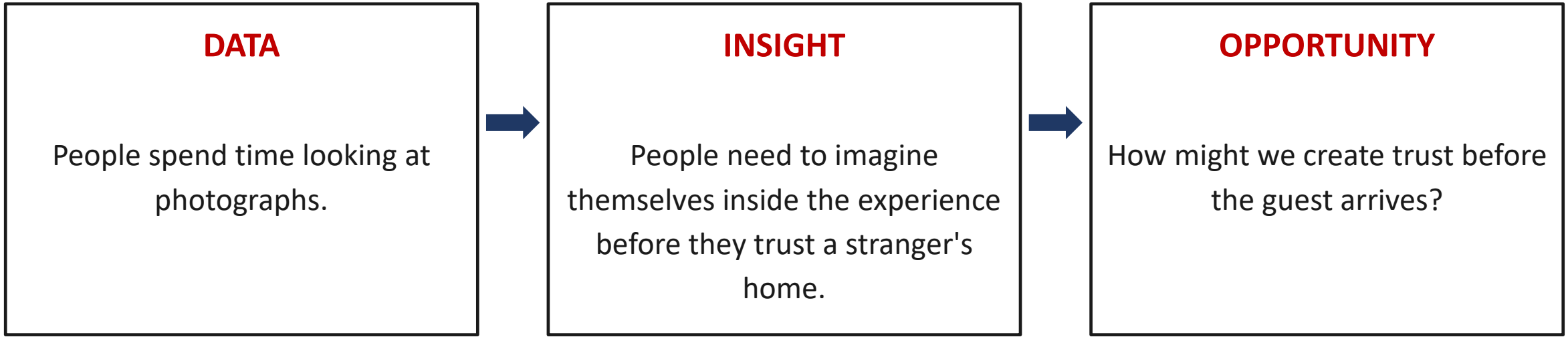
Compare

Find patterns

Find contradictions

Identify opportunities

The Airbnb “Aha!” — From Data to Insight to a New Opportunity




https://youtube.com/shorts/qsw3HX_Fuqs?si=XBLZkhp8mh2Noqwh

Are we asking the right question?

Have we actually seen the problem?

What insight changes how we see the problem?

The first half of innovation is not about generating answers.

It is about learning to ask better questions.

How Do We Create Solutions Nobody Has Thought Of?

Innovation requires two different mental modes:

Imagination and Experimentation.

"Imagine wildly. Build quickly."

How many uses can you think of

for a dead phone ?

EXPECT

Paperweight.

Doorstop.

Weapon.

Place holder

The first five ideas are usually obvious.

The tenth idea starts becoming interesting.

The fiftieth idea may be completely crazy.

Crazy ideas are sometimes where innovation begins.

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  How Do We Make the World Care?

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RUNNING CASES

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Plus: your own team's campus challenge, carried through all three parts.

Step 4 · Generate Ideas

- Meeting 1 — Generate ideas. No criticism allowed.
- Meeting 2 — Kill ideas. Judgment happens later, deliberately.
- Why? If every idea immediately receives criticism — “too expensive,” “not technically possible,” “management won't approve it” — the group stops generating.

RULE

Never generate and judge in the same meeting.

How Might We Provide High-Quality Eye Care to Millions at Affordable Cost?

- A. Build more hospitals.
- B. Develop cheaper surgical robots.
- C. Train more doctors.
- D. Create mobile eye hospitals.
- E. ***Redesign the workflow itself. — particularly interesting.***

THE INTERESTING SHIFT

Not: "How can one doctor work faster?"

But: "How can the entire system remove wasted time and increase access?"

An innovation does not have to be a new object.

It can be a new process, a new business model, a new system architecture,

a new service, or a new combination of existing technologies.

Case: Airbnb — Expanding the Idea Space:

"How Might We Create Trust Between Strangers?"

- Guests needed: affordability, availability, trust, a feeling of safety, confidence.
- Hosts needed: income, simplicity, trust, protection.
- So instead of "build a website," the innovation challenge becomes: "How might we create trust between strangers?"

NOW THE IDEA SPACE EXPANDS TO

Identity verification
Reviews
Photographs
Ratings
Secure payments
Profiles
Messaging
Guarantees

https://youtu.be/16cM-RFid9U?si=p3IwqkURNirZncnD



Step 5 · Make Ideas Tangible

- Low fidelity: a sketch, a cardboard model, a storyboard, a paper interface, a role play.
- Medium fidelity: a clickable prototype, a 3D print, a functional mock-up.
- High fidelity: a working MVP.

THE OBJECTIVE

Don't prototype to impress.

Prototype to learn.

Case: Airbnb

- Imagine the founders had spent ₹50 crore equivalent building a perfect global platform before discovering what users wanted. That would have been dangerous.
- Instead, the early story of Airbnb demonstrates a startup learning directly from real users and iterating its offering based on what people actually experienced.

THE LESSON

Build something.

The 10-Minute IIT Madras Design Challenge

Today's challenge: "How might we make everyday movement across a large campus effortless, sustainable and inclusive?"

Time	What to do
Min 1–3	Generate 20 ideas. Rule: no criticism.
Min 4	Choose the most interesting idea.
Min 5–8	Create a prototype — a paper app, a map, a physical model, or a service storyboard.
Min 9–10	One team member demonstrates it — to your own pod, not the whole hall.

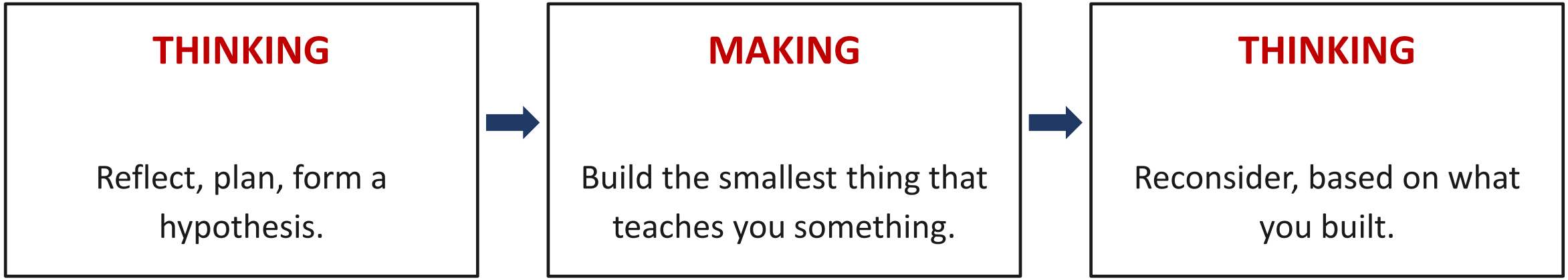
How many teams changed their idea while building?

Exactly.

You did not first have a perfect idea and then make it.

Making the idea helped create the idea.

Thinking → Making → Thinking



Not: Thinking → Thinking → Thinking → Perfect Answer

The best innovators do not wait for certainty before making something. They make something to reduce uncertainty.

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How Do We Know It Works — and How Do We Make the World Care?

The prototype is not the end.

The customer is the experiment.

"Watch people use it. Learn. Tell the world why it matters."

How many of you have worked for months on a project

and then discovered that nobody actually wanted it?

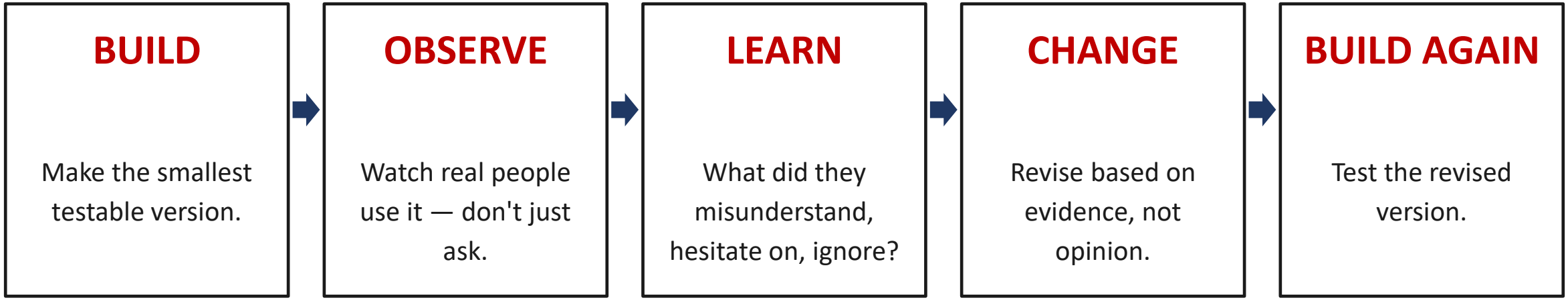
THE POINT

The purpose of testing is not to prove that you are right. It is to discover how you might be wrong.

Step 6 · Test to Learn — Testing Should Answer Four Questions

Dimension	The Question
Desirability	Do people actually want this?
Usability	Can people actually use it?
Feasibility	Can we build it?
Viability	Can the solution sustain itself economically?

The Testing Loop — Build → Observe → Learn → Change → Build Again



The critical word is OBSERVE. Do not simply ask “Do you like my invention?” — people often say “Yes, very nice!” and then never use it.

Case: Airbnb — Your First Solution Is a Hypothesis. Not a Conclusion.

- Airbnb's development illustrates the importance of learning from actual user behaviour rather than treating the founders' assumptions as truth.
- The company's evolution required attention to the real experience of both guests and hosts.
- A startup is essentially saying: “We believe this group of people has this problem, and we believe this solution will create value.”
- Testing asks: how do we know?

WHY THIS MATTERS FOR STARTUPS

Airbnb's early turnaround story is frequently used as a Design Thinking case

because the team sought direct customer understanding and altered the offering accordingly.

Case: Aravind — Testing a Large-Scale Service System Is Not Only "Does the Technology Work?"

- Does the patient understand the process?
- Can they access the service?
- Where does waiting occur?
- What prevents people from seeking treatment?
- Does affordability change adoption?
- Can quality remain high at scale?

CRITICAL FOR ENGINEERS

Technical validation is not the same as user validation.

Technical Validation ≠ User Validation —

Your Electric Vehicle May Work Perfectly. But…

- Can the customer charge it?
- Does the range create anxiety?
- Does the interface confuse them?
- Can it be serviced?
- Can they afford it?

ENGINEERING ASKS

Does it work?

Innovation requires adoption.

And adoption requires understanding.

Step #7 - The Story Formula

Element	Ask Yourself
The Person	Who is experiencing the problem?
The Struggle	What frustrates or limits them?
The Insight	What did we discover?
The Idea	What did we create?
The Evidence	What did we learn through testing?
The Impact	What changes if this succeeds?

Back To The Loop

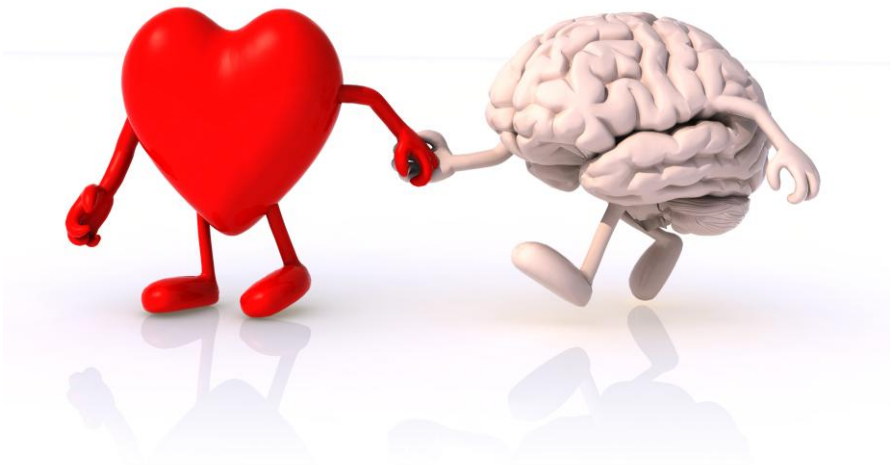
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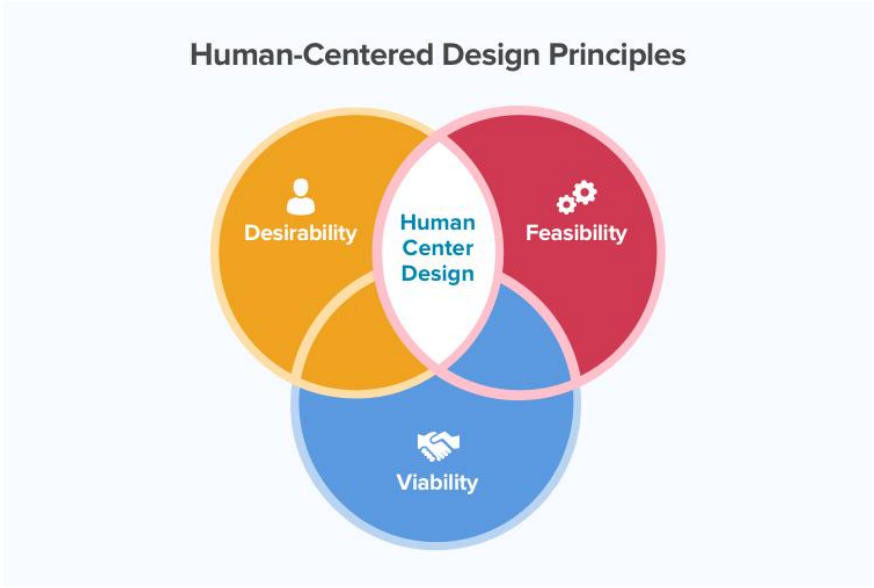
Every team will revisit these seven steps out of order, more than once.

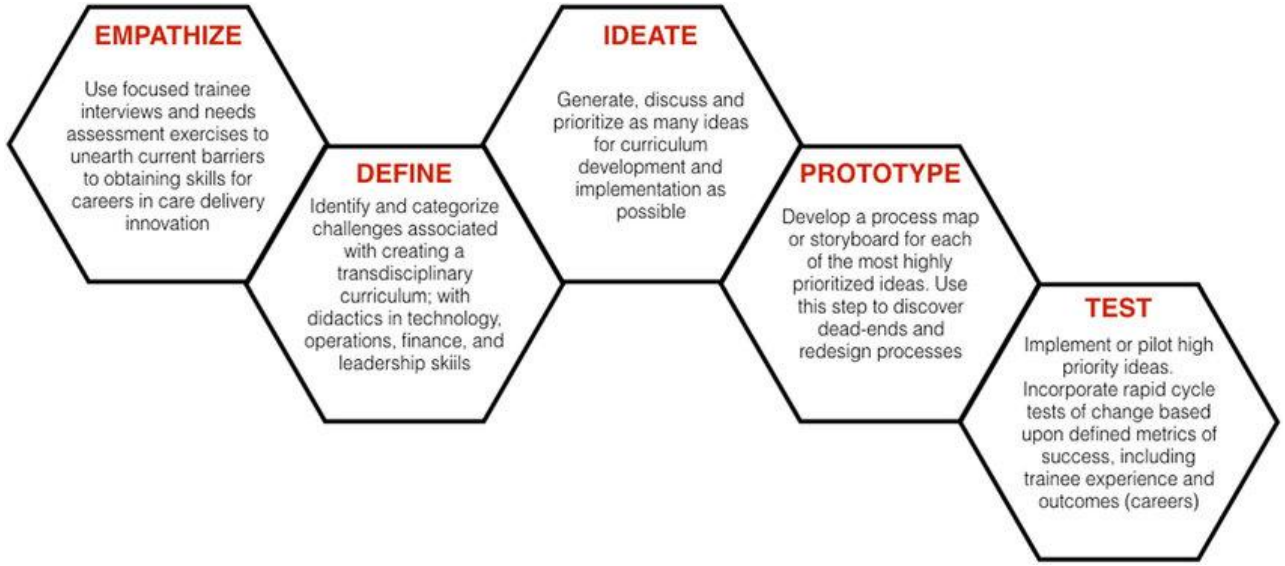
That is not a failure of process — it is the process.

A Design Thinking Mindset

- Empathy – Put People First
- Curiosity – Ask Open Questions
- Bias to Action – Make Ideas Real – Build Protos quick
- Embrace Ambiguity – Don’t fear the Unknown – Stay open to change
- Learn from Failure – Treat Mistakes as Data
- Radical Collaboration – Work with diverse groups




Human-Centered Design Principles
Desirability
Human Center Design
Feasibility
Viability


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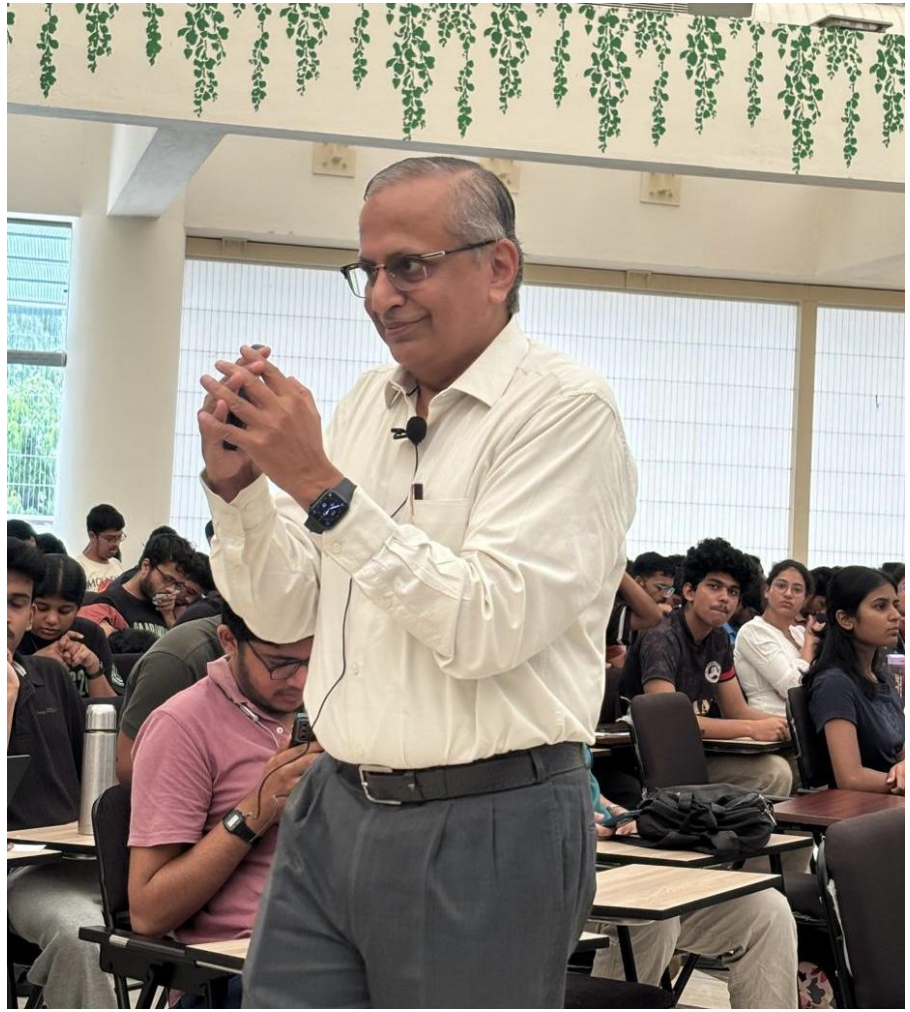
Technology gives us the power to create almost anything.

Design Thinking gives us the discipline to ask whether we should —

and whether it matters to people.

Ask better questions. · Build to learn. · Make people care.

Dr Raghuttama Rao, Chief Executive Officer
Gopalakrishnan-Deshpande Centre for Innovation & Entrepreneurship, IIT Madras



https://gdciitm.org/

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